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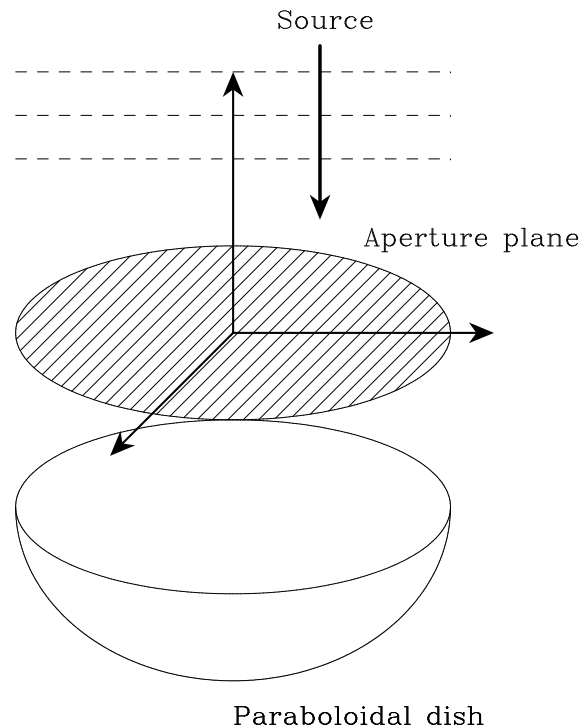
# Lecture 5: Filled Aperture Antennas: Power Patterns

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# A Circular Aperture

The term **aperture** refers to the opening through which all rays pass. For example, the aperture of a paraboloidal reflector antenna would be the plane circle, normal to the rays from a distant point source, that just covers the paraboloid. The phase of the plane wave from a distant point source would be constant across the aperture plane when the aperture is perpendicular to the line-of-sight.



The area of the aperture is  $\pi r^2$  where  $r$  is the radius.

# A Rectangular Aperture

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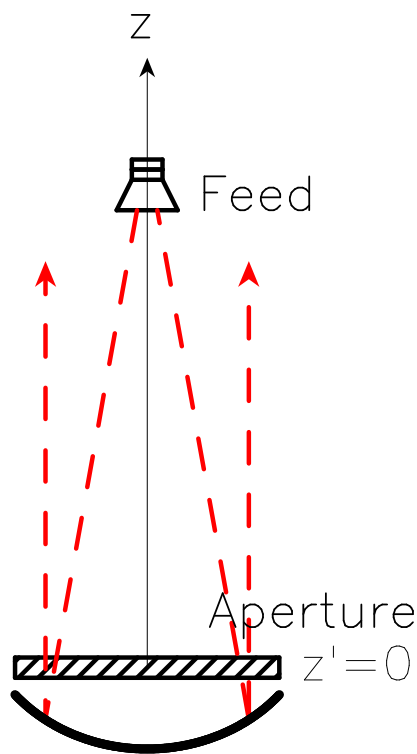


Another example of an aperture is **the mouth of a waveguide horn antenna**. “Doc” Ewen looking into the rectangular aperture of the horn antenna used to discover the  $\lambda = 21$  cm line of neutral hydrogen. Note waveguide horn antennas are used for long  $\lambda$ .

The area of the aperture is  $L_x L_y$  where  $L_x$  and  $L_y$  are the lengths of the two sides.

# Power Pattern of an Aperture

Reciprocity: The Receiving Power Pattern is the same as the Transmitting Power Pattern.



For a paraboloidal dish, let's assume that a feed is installed above it. The feed illuminates the aperture with a sine wave of fixed frequency  $\nu = \omega/(2\pi)$  and electric field strength  $g(\mathbf{x})$  (illumination or grading function) that varies across the aperture. The illumination induces currents in the reflector (dish) but can be considered in the aperture view from afar. The current densities  $\mathbf{J}$  will vary with both position and time:

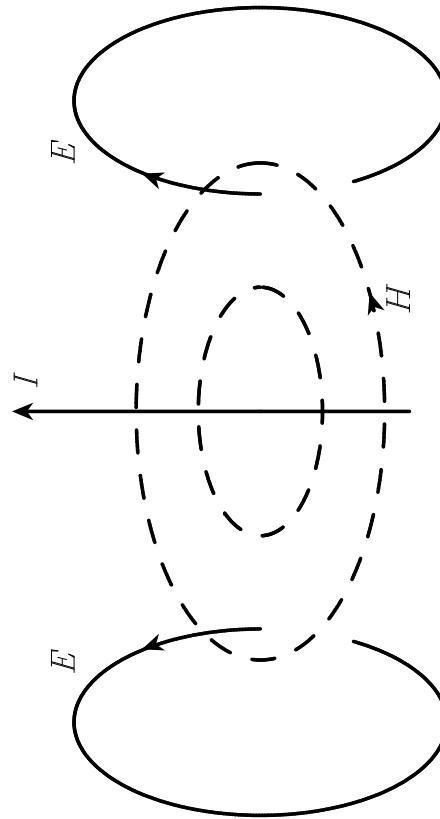
$$\mathbf{J} \propto g(\mathbf{x}) \exp(-i\omega t)$$

The aperture can be treated as a collection of small elements which act individually as small antennas.

The electric field produced by the whole aperture at large distances is just the vector sum of the elemental electric fields from these small antennas (see next few slides)

# EM radiation induced by a current

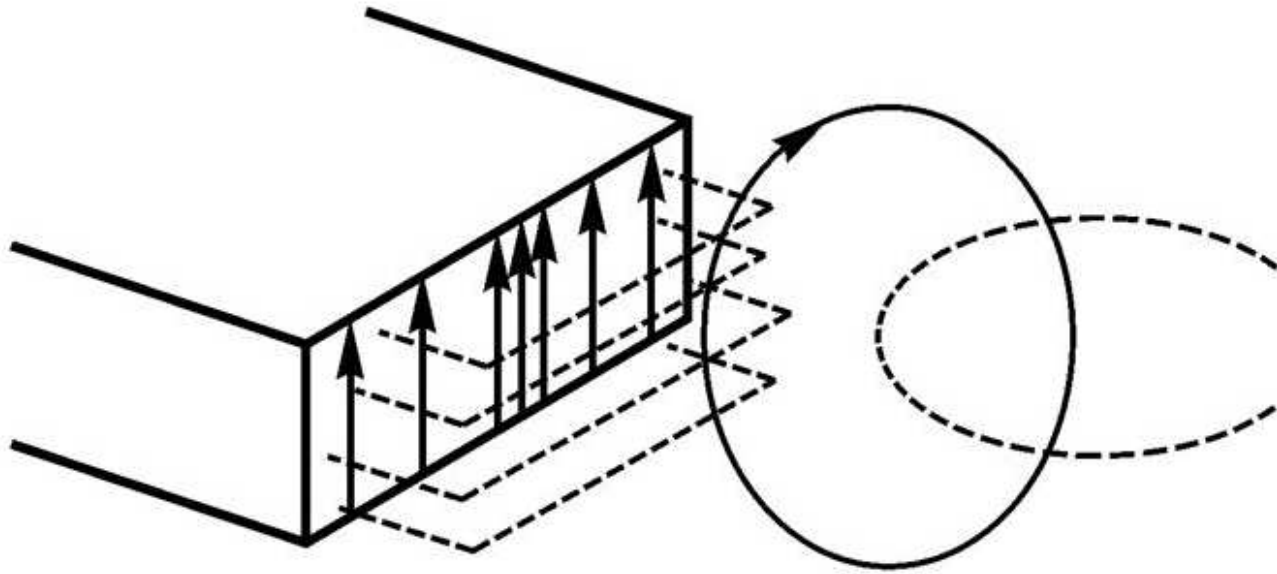
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A temporal variation of current in the aperture can produce a EM wave propagating away. A steady current can induce a  $\mathbf{H}$  field around it. As the current changes, the  $\mathbf{H}$  field changes, inducing a  $\mathbf{E}$  field and thus producing a wave propagating away.

# EM radiation from a waveguide

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The open face of a waveguide will radiate into the free space. A change of  $\mathbf{E}$  field at the opening of the waveguide induces a change of  $\mathbf{H}$  outside the waveguide. This change of  $\mathbf{H}$  field will cause a change of  $\mathbf{E}$  field, thus producing a wave propagating away.

# Radiation field of Localized Sources

How to calculate the radiation field produced by a localized current such as that in Hertz dipole and general antennas? Usually, the current can be decoupled as in Hertz dipole,

$$\mathbf{J}(\mathbf{x}, t) = \mathbf{J}(\mathbf{x})e^{-i\omega t}$$

The amplitude  $\mathbf{J}(\mathbf{x})$  can be a complex quantity, so that the phases of the oscillation will depend on the position  $\mathbf{x}$ . Anyway, the vector potential  $\mathbf{A}$  generated by this current is

$$\mathbf{A}(\mathbf{x}, t) = \frac{\mu}{c} \iiint_V \frac{\mathbf{J}(\mathbf{x}', t - \frac{|\mathbf{x} - \mathbf{x}'|}{v})}{|\mathbf{x} - \mathbf{x}'|} d^3x' = \frac{\mu}{c} \iiint_V \frac{\mathbf{J}(\mathbf{x}')e^{-i\omega(t - \frac{|\mathbf{x} - \mathbf{x}'|}{v})}}{|\mathbf{x} - \mathbf{x}'|} d^3x' \equiv \mathbf{A}(\mathbf{x})e^{-i\omega t}$$

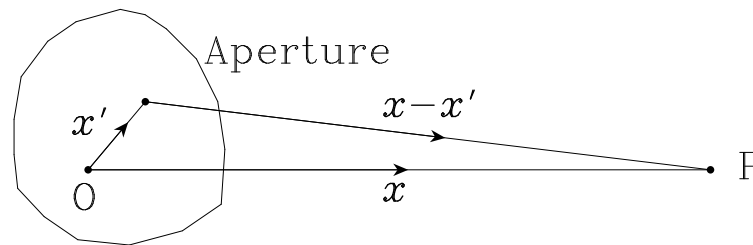
with  $v = \omega/k$  and

$$\mathbf{A}(\mathbf{x}) = \frac{\mu}{c} \iiint_V \mathbf{J}(\mathbf{x}') \frac{e^{ik|\mathbf{x} - \mathbf{x}'|}}{|\mathbf{x} - \mathbf{x}'|} d^3x'$$

$\therefore \mathbf{A} \Rightarrow \mathbf{B} \Rightarrow \mathbf{E}$ ,  $\Rightarrow$  the electric field produced by the whole aperture at large distances is just the vector sum of the elemental electric fields from these small antennas. The exponential part is just the phase for the interference.

# Fraunhofer far-field approximation

For far field, the vector potential can be solved with some approximation. e.g.,



**Fraunhofer approximation:** assuming the linear dimensions of the antenna structures are small compared to the distance of the point  $\mathbf{x}$ , i.e.,  $\mathbf{x} \gg \mathbf{x}'$ . Let  $\mathbf{n} = \mathbf{x}/|\mathbf{x}|$  be a unit vector in the direction of  $\mathbf{x}$ ,  $r = |\mathbf{x}|$ , and  $\mathbf{m}$  a unit vector normal to  $\mathbf{x}$ , then we have

$$|\mathbf{x} - \mathbf{x}'| = |r\mathbf{n} - \mathbf{x}'| = \sqrt{(r - \mathbf{n} \cdot \mathbf{x}')^2 + (\mathbf{m} \cdot \mathbf{x}')^2} \simeq r - \mathbf{n} \cdot \mathbf{x}'$$

$\Rightarrow$  only need to consider the projection of  $\mathbf{x}'$  on  $\mathbf{x}$ . For  $r \gg \mathbf{n} \cdot \mathbf{x}'$ , we have

$$\mathbf{A}(\mathbf{x}) = \frac{\mu}{c} \iiint_V \mathbf{J}(\mathbf{x}') \frac{e^{ik|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x} - \mathbf{x}'|} d^3x' \approx \frac{\mu}{c} \frac{e^{ikr}}{r} \iiint_V \mathbf{J}(\mathbf{x}') e^{-ik\mathbf{n} \cdot \mathbf{x}'} d^3x'$$

$\Rightarrow$  Huygen's Principle: Each point  $\mathbf{x}'$  produces an expanding spherical wave for far field. The total is the vector sum of waves from all the points.

# EM Radiation

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We then can calculate the  $\mathbf{B}$  field with

$$\mathbf{B} = \nabla \times \mathbf{A}$$

As for the  $\mathbf{E}$  field, from the Ampere-Maxwell's law, we have

$$\dot{\mathbf{D}} = \epsilon \dot{\mathbf{E}} = c \nabla \times \mathbf{H} - 4\pi \mathbf{J}$$

In free space, far from the antenna,  $\sigma = 0$  and thus  $\mathbf{J} = 0$ . Also  $\epsilon = 1$ ,  $\mu = 1$ . So,

$$\dot{\mathbf{E}} = c \nabla \times \mathbf{B}$$

For a harmonic wave  $\mathbf{E} \propto e^{-i\omega t}$ , we have  $\dot{\mathbf{E}} = -i\omega \mathbf{E}$ . Thus

$$\mathbf{E} = \frac{ic}{\omega} \nabla \times \mathbf{B} = \frac{i}{k} \nabla \times (\nabla \times \mathbf{A})$$

with  $c = \omega/k$ .

# Power Pattern of an Aperture

Let the aperture be in the plane  $z' = 0$ . For simplicity, the current in the aperture is assumed to be parallel (i.e., one polarization) (the current is actually parallel to dish surface), with the unit vector of the linear current density (esu s<sup>-1</sup> cm<sup>-1</sup>)  $\mathbf{J}_0 = (0, J_0, 0)$  (in y-direction). The current in a surface element  $dx' dy'$  at  $\mathbf{x}'$  in the aperture  $A$  is then

$$I = J_0 g(\mathbf{x}') dx'$$

where  $g(\mathbf{x}')$  is the illumination or grading function. Remember the  $\mathbf{E}$  field for Hertz antenna

$$E_\theta = -i \sqrt{\frac{\mu}{\epsilon}} \frac{k I \Delta l \sin \theta}{cr} \exp[-i(\omega t - kr)]$$

If  $\mathbf{x}$  is approximately perpendicular to  $A$ , for example, the only component of the  $\mathbf{E}$  field by this current element in  $\mathbf{x}$  is (with  $\epsilon = 1$ ,  $\mu = 1$ ,  $k = \frac{2\pi}{\lambda}$ ,  $r = |\mathbf{x} - \mathbf{x}'|$ ,  $E_\theta \Rightarrow dE_y$ , and  $\Delta l \Rightarrow dy'$ )

$$dE_y = -\frac{i2\pi}{c} \lambda J_0 g(\mathbf{x}') \frac{F_e(\mathbf{n})}{|\mathbf{x} - \mathbf{x}'|} e^{-i(\omega t - k|\mathbf{x} - \mathbf{x}'|)} \frac{dx'}{\lambda} \frac{dy'}{\lambda}$$

$F_e(\mathbf{n})$  is the field pattern of the current element for the direction  $\mathbf{n} = \mathbf{x}/|\mathbf{x}|$  which, for the Hertz dipole is  $\sin \theta$  where  $\theta$  is the angle betw.  $\mathbf{n}$  and  $\mathbf{J}_0$ . (Remember that the  $\mathbf{E}$  field peaks at  $\theta = 90^\circ$ ).

# Far field

For far field,  $r \gg |\mathbf{n} \cdot \mathbf{x}'|$ , we can neglect  $\mathbf{n} \cdot \mathbf{x}'$  except in the exponent. Thus,

$$\begin{aligned} E_y &= -\frac{i2\pi}{c} \lambda J_0 \frac{F_e(\mathbf{n})}{r} e^{-i(\omega t - kr)} \iint g(\mathbf{x}') e^{-ik\mathbf{n} \cdot \mathbf{x}'} \frac{dx'}{\lambda} \frac{dy'}{\lambda} \\ &\equiv -\frac{i4\pi^2}{c} \lambda J_0 \frac{F_e(\mathbf{n})}{r} f(\mathbf{n}) e^{-i(\omega t - kr)} \end{aligned}$$

with a 2D Fourier integral of the illumination function

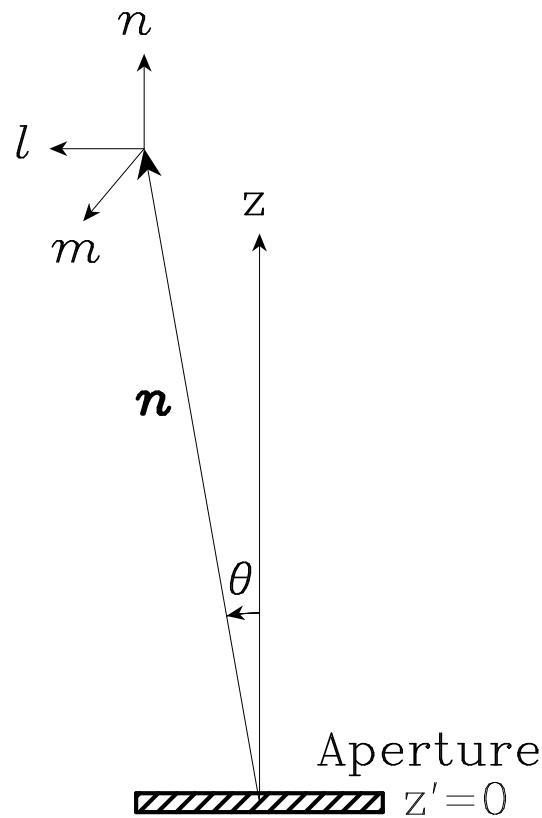
$$f(\mathbf{n}) = \frac{1}{2\pi} \iint g(\mathbf{x}') e^{-ik\mathbf{n} \cdot \mathbf{x}'} \frac{dx'}{\lambda} \frac{dy'}{\lambda}$$

In the far field, the electric-field pattern of an aperture antenna is the Fourier transform of the electric field illuminating the aperture. Note that this soln can be generalized for any plane aperture with parallel current (one polarization). Moreover, even though our soln are derived assuming  $\mathbf{x}$  is approximately perpendicular to  $A$ , the soln is actually for all direction. A similar expression for the magnetic field strength  $\mathbf{H}$ . The normalized power pattern is

$$P_n(\mathbf{n}) = \frac{P(\mathbf{n})}{P_{\max}} = \frac{|\mathbf{E} \cdot \mathbf{E}'|}{|\mathbf{E} \cdot \mathbf{E}'|_{\max}} = \frac{|f(\mathbf{n})|^2}{|f_{\max}|^2}$$

# Example: Rectangular Aperture

**Uniform illumination:** the strength of the illumination is independent of position across the aperture:



$$g(x, y) = \begin{cases} 1 & \text{for } |x| \leq L_x/2, |y| \leq L_y/2 \\ 0 & \text{otherwise} \end{cases}$$

Let  $a$  be the angle between  $\mathbf{n}$  and  $\hat{x}$ ,  $b$  the angle between  $\mathbf{n}$  and  $\hat{y}$ , and  $\theta$  the angle between  $\mathbf{n}$  and  $\hat{z}$ . Then the direction cosines are equivalent to the  $(x, y, z)$  coordinates of a unit vector  $\mathbf{n}$ ,

$$\begin{aligned} l &= \cos a = \mathbf{n} \cdot \hat{x} \\ m &= \cos b = \mathbf{n} \cdot \hat{y} \\ n &= \cos \theta = \mathbf{n} \cdot \hat{z} \end{aligned}$$

with  $\mathbf{n} = (l, m, n)$ ,  $l^2 + m^2 + n^2 = 1$ , and  $l^2 + m^2 = \sin^2 \theta$ .

Thus,  $\mathbf{n} \cdot \mathbf{x}' = (l, m, n) \cdot (x', y', 0) = lx' + my'$

What we care is the angle from the  $z$ -axis, so let  $\theta_x = \frac{\pi}{2} - a$  and  $\theta_y = \frac{\pi}{2} - b$ , then  $l = \sin \theta_x$  and  $m = \sin \theta_y$ , where  $\theta_x$  is now the angle from the  $(y, z)$  plane and  $\theta_y$  the angle from the  $(x, z)$  plane. And  $\sin^2 \theta = l^2 + m^2 = \sin^2 \theta_x + \sin^2 \theta_y$  and  $n = \cos \theta$ .

# Example: Rectangular Aperture

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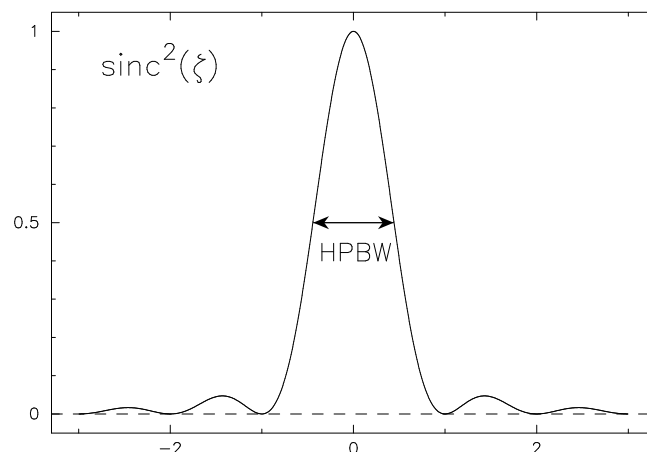
Taking the Fourier Transform of the illumination function, we have

$$\begin{aligned} f(l, m) &= \frac{1}{2\pi} \int \int g(x', y') \exp\left\{-i \frac{2\pi}{\lambda} (lx' + my')\right\} \frac{dx'}{\lambda} \frac{dy'}{\lambda} \\ &= \frac{1}{2\pi} \int_{-L_x/2}^{L_x/2} \int_{-L_y/2}^{L_y/2} \cos\left(\frac{2\pi}{\lambda} lx'\right) \cos\left(\frac{2\pi}{\lambda} my'\right) \frac{dx'}{\lambda} \frac{dy'}{\lambda} \\ &\propto \frac{\sin(\pi l L_x / \lambda)}{\pi l L_x / \lambda} \frac{\sin(\pi m L_y / \lambda)}{\pi m L_y / \lambda} \\ &= \operatorname{sinc}\left(\frac{l L_x}{\lambda}\right) \operatorname{sinc}\left(\frac{m L_y}{\lambda}\right) \leq 1 \end{aligned}$$

Here  $\operatorname{sinc}(x) \equiv \frac{\sin(\pi x)}{\pi x}$ . Thus, the normalized power pattern,

$$P_n(l, m) = \left[ \operatorname{sinc}\left(\frac{l L_x}{\lambda}\right) \operatorname{sinc}\left(\frac{m L_y}{\lambda}\right) \right]^2$$

# Power Pattern



with  $\xi = lL_x/\lambda$ . The main beam is the solid angle betw. the first nulls of  $P_n$  at

$$l_0 = \pm\lambda/L_x \qquad m_0 = \pm\lambda/L_y$$

and the half power beam width (HPBW), i.e., the angle betw. those points of the main beam for which  $P_n = 1/2$  at (e.g., from the  $(y, z)$  plane)

$$l = \sin \theta_x = 0.44\lambda/L_x$$

is ( $\sin \theta_x \sim \theta_x$  if  $L_x \gg \lambda$ ),

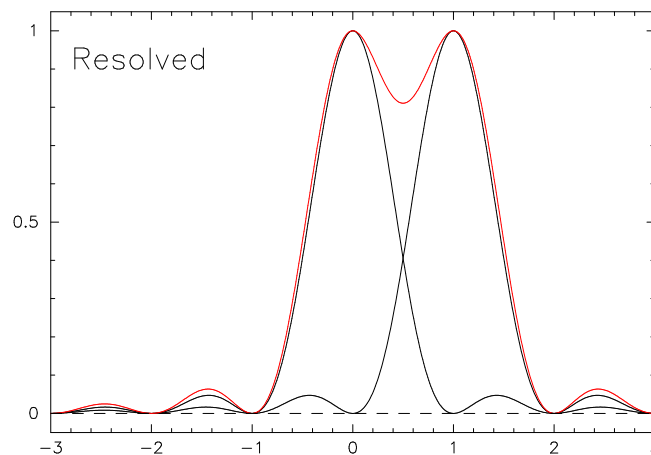
$$\text{HPBW}_x = 0.88\lambda/L_x \text{ rad} = 50.3^\circ \lambda/L_x$$

# Resolving Power

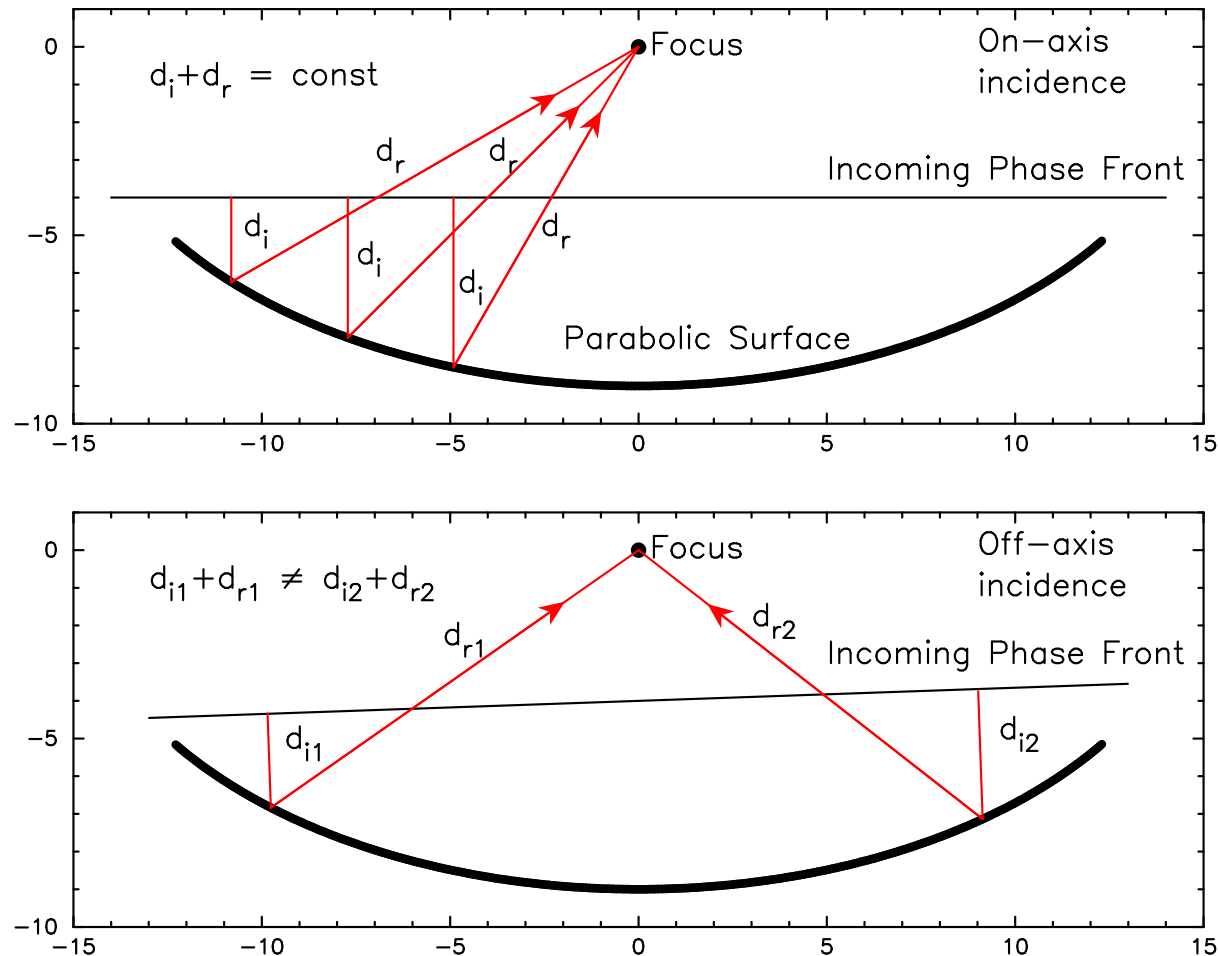
The reciprocity theorem says that this analysis of the transmitting power pattern also yields the receiving power pattern. In receiving terms, the power pattern represents the point-source response, since the point source can be considered as the source of a plane wave receiving in the aperture plane.

For a uniformly illuminated aperture, scanning a radio telescope beam across a point source will cause  $T_A$  to vary as  $\text{sinc}^2$ , and the half-power response width will equal the transmitting HPBW.

The receiving HPBW is sometimes called the resolving power of a telescope because two equal point sources separated by the HPBW can just be resolved by the Rayleigh criterion that the peak response to one source coincides with the first minimum response to the other, so the total response has a slight minimum midway between the point sources.



# Power Pattern: Origin



An antenna's response is a result of phase summation at the focus. First null will occur at the angle where the extra distance for a wave at the center of antenna is in anti-phase with that from edge.

# Beam Solid Angle + Directive gain

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With the normalized power pattern, we can derive the beam solid angle

$$\Omega_A = \int \int P_n(l, m) dl dm$$

Here,  $l = \sin \theta_x = \xi \frac{\lambda}{L_x}$ . Since the side lobes become insignificant for  $\xi > 10$  or  $l > 10 \frac{\lambda}{L_x}$ , for  $L_x > 10\lambda$ , it suffices to have  $l = [-1, 1]$  (i.e. [the front direction of the antenna](#)) and

$$\Omega_A = \int_{-1}^{+1} \int_{-1}^{+1} P_n(l, m) dl dm = \int_{-1}^{+1} \int_{-1}^{+1} \left[ \text{sinc}\left(\frac{lL_x}{\lambda}\right) \text{sinc}\left(\frac{mL_y}{\lambda}\right) \right]^2 dl dm = \frac{\lambda^2}{L_x L_y}$$

Note  $\int_{-\infty}^{\infty} \text{sinc}^2(x) dx = 1$ . Thus, the directive gain  $G$  in any direction is

$$G(l, m) = \frac{4\pi}{\Omega_A} P_n(l, m) = \frac{4\pi L_x L_y}{\lambda^2} P_n(l, m)$$

# Directivity + Aperture Efficiency

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Thus the directivity is

$$D = G_{\max} = \frac{4\pi L_x L_y}{\lambda^2}$$

and the effective aperture is

$$A_e = \frac{D\lambda^2}{4\pi} = L_x L_y$$

Therefore, the aperture efficiency is

$$\eta_A = \frac{A_e}{A_g} = 1$$

Thus  $\eta_A = 1$  for an ideal uniformly illuminated aperture and  $\eta_A < 1$  otherwise.

The main beam efficiency is

$$\eta_B = \frac{\Omega_{\text{MB}}}{\Omega_A} = \frac{\iint_{\text{MB}} P_n(l, m) dl dm}{\Omega_A} \approx 0.8$$

# Long waveguide horns

Long waveguide horns are nearly uniformly illuminated unblocked apertures, so their actual gains and effective collecting areas can be calculated accurately. They can therefore be used to measure the absolute flux densities of strong sources such as Cas A and Cyg A and define the flux-density scales used by radio astronomers (see Baars et al. 1977, A&A, 61, 99).

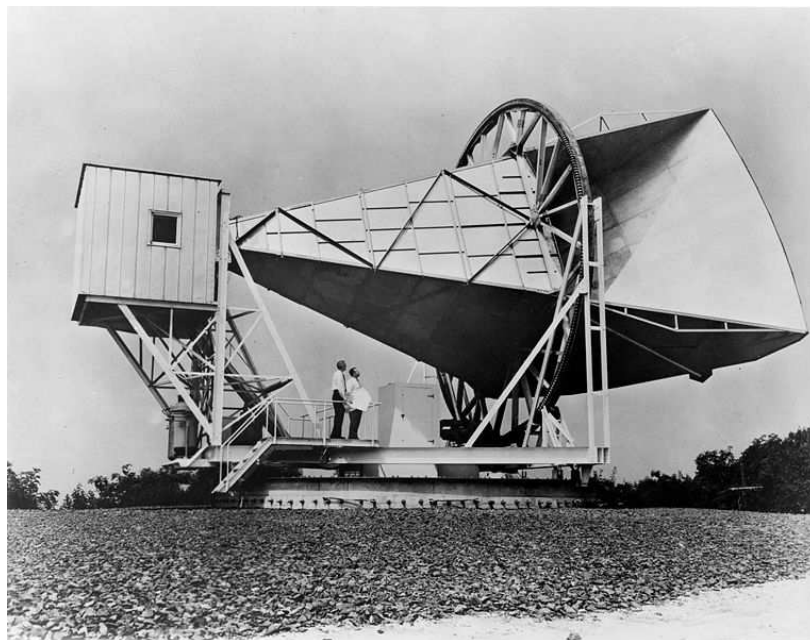


FIGURE 17. A side view of the horn that will be used for precision calibration measurements of the radio source Cas A. The horn is 120 feet long and the rectangular cross-section is 13x17 feet at the upper end. The horn is fixed with an orientation such that once each day the rotation of the earth sweeps the axis of the horn across Cas A.

The “Little Big Horn” at Green Bank, WV in 1959. It was used to measure the absolute flux density of the strong source Cas A. An excess noise of  $T \approx 3.5$  K was found but not recognized as important.

# Horn Antenna

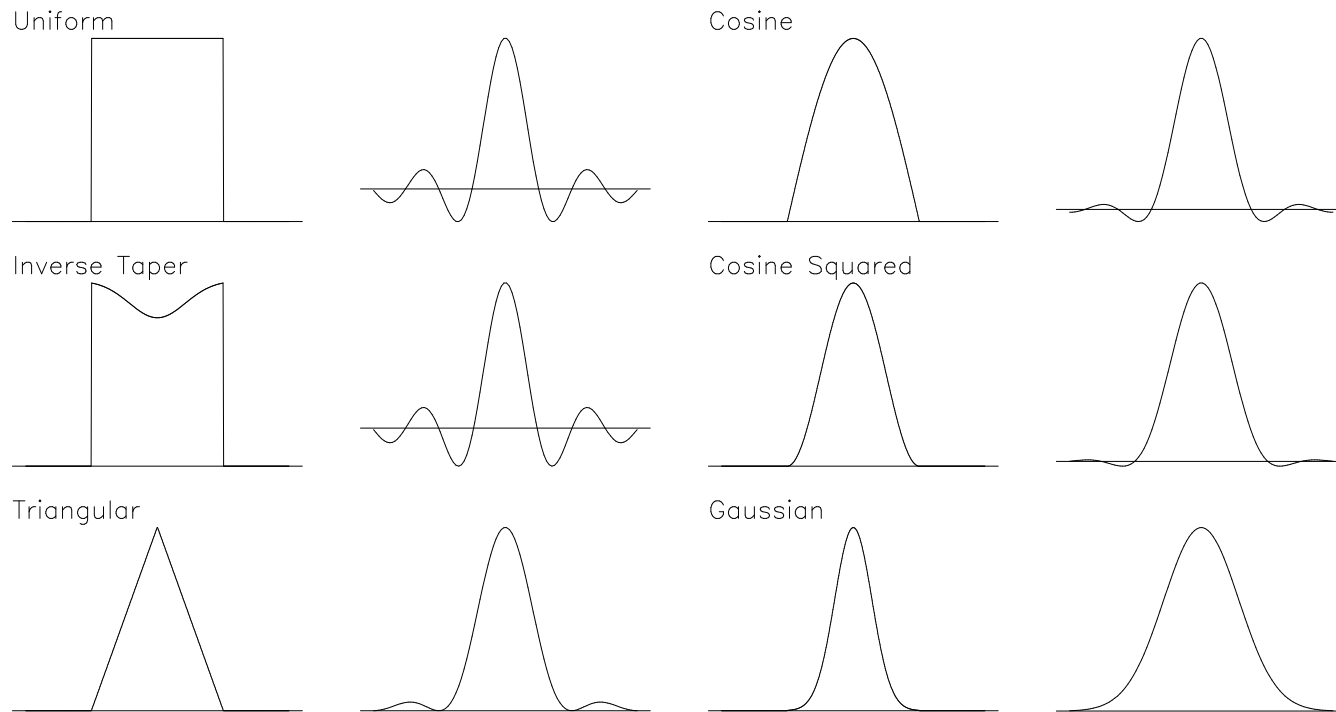
In 1964, David Wilkinson and Peter Roll, Dicke's colleagues at Princeton, began constructing a Dicke radiometer to measure the CMB. In 1965, Arno Penzias and Robert Wilson at the Crawford Hill location of Bell Telephone Lab. in nearby Holmdel Township had built a Dicke radiometer that they intended to use for radio astronomy and satellite communication experiments. Their instrument had an excess 3.5 K in  $T_A$  which they could not account for. After receiving a telephone call from Crawford Hill, Dicke at Princeton turned to his group: "Boys, we've been scooped." A meeting between the Princeton and Crawford Hill groups determined that the antenna temperature was indeed due to the CMB. Penzias and Wilson received the 1978 Nobel Prize in Physics for their discovery.



The Horn Antenna on which Penzias and Wilson discovered the cosmic microwave background (CMB).

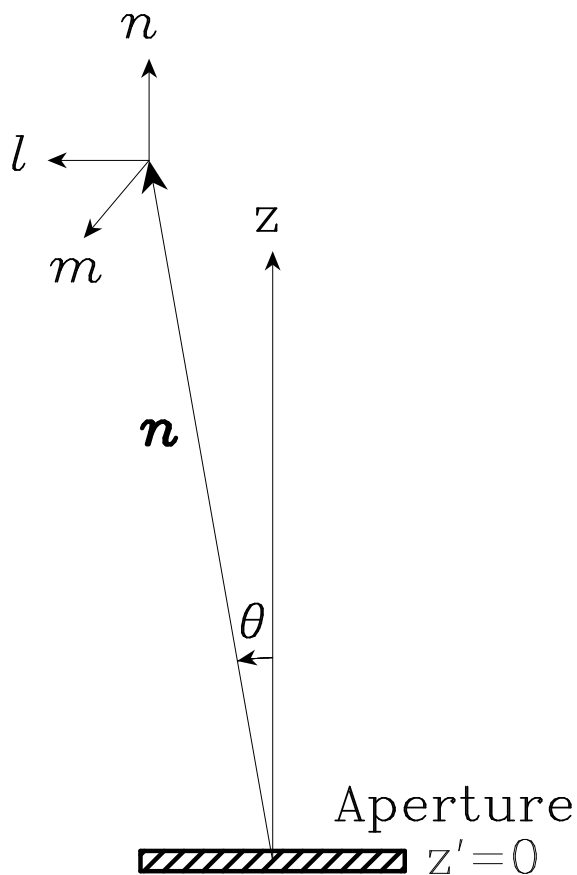
# Examples of 1D Power Pattern

Some examples of current gradings ( $g$ ) and corresponding electric far field pattern  $E$ :



Note that The Fourier Transform of a Gaussian is another Gaussian. The sharper decrease in the gradings at the edge, the larger the sidelobe.

# For Circular Aperture



$$f(\mathbf{n}) = \frac{1}{2\pi} \iint g(\mathbf{x}') e^{-ik\mathbf{n} \cdot \mathbf{x}'} \frac{dx'}{\lambda} \frac{dy'}{\lambda}$$

For a circular aperture, polar coordinates  $(\varrho, \varphi)$  is convenient:

$$x = \lambda \varrho \cos \varphi$$

$$y = \lambda \varrho \sin \varphi$$

with  $\varrho$  in unit of  $\lambda$  and the aperture defined by  $\varrho \leq D/2\lambda$ .

If the current grading  $g$  depends on

$\varrho$  only, the resulting beam pattern will also show **circular symmetry**.

So, only a single value,  $u = \sin \theta$ , the sine of the angle between  $\mathbf{n}$  and the direction of the main beam (assumed to be in the z-direction) is needed. The  $\mathbf{E}$  field is proportional to (with  $k\mathbf{n} \cdot \mathbf{x}' = 2\pi u \varrho \cos \varphi$ )

$$f(u) = \frac{1}{2\pi} \int_0^{2\pi} \int_0^\infty g(\varrho) e^{-i2\pi u \varrho \cos \varphi} \varrho d\varrho d\varphi$$

# Circular Aperture: Power Pattern

The  $\mathbf{E}$  field is proportional to

$$\begin{aligned} f(u) &= \frac{1}{2\pi} \int_0^{2\pi} \int_0^\infty g(\varrho) e^{-i2\pi u \varrho \cos \varphi} \varrho d\varrho d\varphi \\ &= \int_0^\infty g(\varrho) \frac{1}{2\pi} \int_0^{2\pi} e^{-i2\pi u \varrho \cos \varphi} d\varphi \varrho d\varrho \end{aligned}$$

which can be written as [Hankel transform of the current grading](#) as

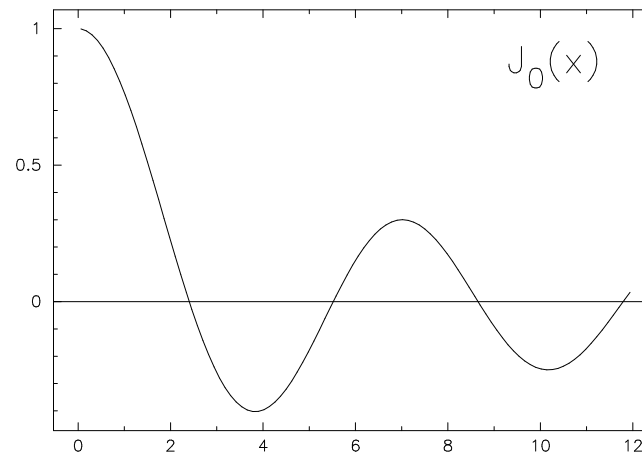
$$f(u) = \int_0^\infty g(\varrho) J_0(2\pi u \varrho) \varrho d\varrho$$

with the [Bessel function](#) of order zero (with  $q \Rightarrow 2\pi u \varrho$ )

$$J_0(q) = \frac{1}{2\pi} \int_0^{2\pi} e^{iq \cos \varphi} d\varphi \quad ; \quad J_0(q) = J_0(-q)$$

# Circular Aperture: Power Pattern

$$J_0(q) = \frac{1}{2\pi} \int_0^{2\pi} e^{iq \cos \varphi} d\varphi \quad ; \quad J_0(q) = J_0(-q)$$



The normalized beam pattern is ( $\because J_0(0) = 1$ )

$$P_n(u) = \left[ \frac{\int_0^\infty g(\varrho) J_0(2\pi u \varrho) \varrho d\varrho}{\int_0^\infty g(\varrho) \varrho d\varrho} \right]^2$$

# Uniformly illuminated circular apert.

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$$g(\varrho) = \begin{cases} 1 & \text{for } \varrho \leq D/2\lambda \\ 0 & \text{otherwise} \end{cases}$$

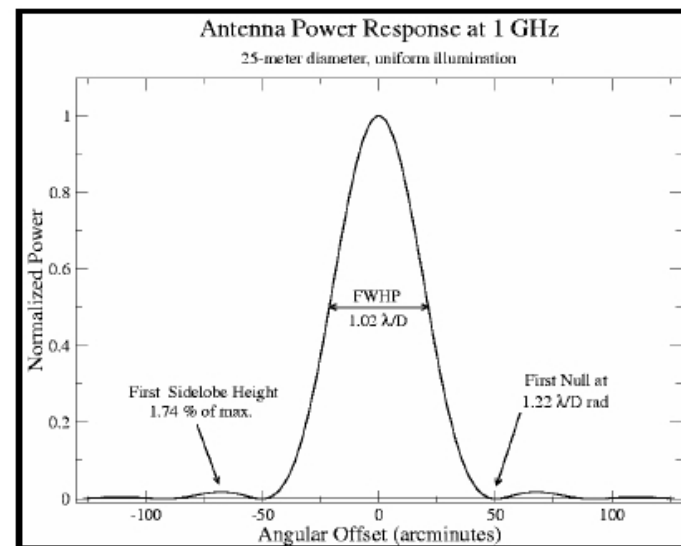
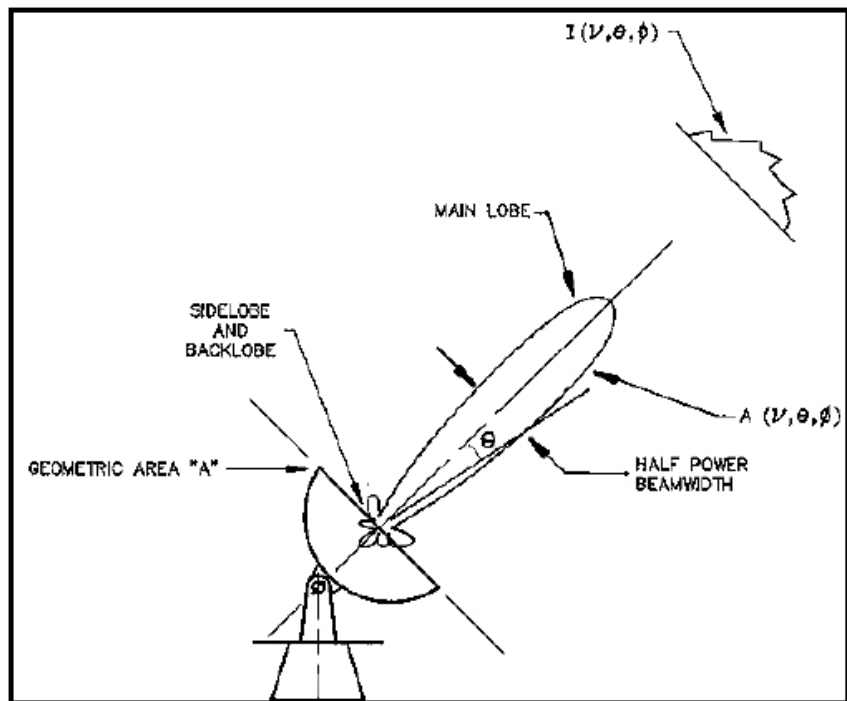
Then the power pattern is

$$\begin{aligned} P_n(u) &= \left[ \frac{\int_0^{D/2\lambda} J_0(2\pi u \varrho) \varrho d\varrho}{\int_0^{D/2\lambda} \varrho d\varrho} \right]^2 \\ &= \left[ \frac{2}{(\pi u D / \lambda)^2} \int_0^{\pi u D / \lambda} J_0(q) q dq \right]^2 && \text{with } q = 2\pi u \varrho \\ &= \left[ \frac{2J_1(\pi u D \lambda)}{\pi u D / \lambda} \right]^2 && \text{with } J_1(x) = \frac{\int_0^x x J_0(x) dx}{x} \\ &= \Lambda_1^2(\pi u D / \lambda) \end{aligned}$$

with

$$\Lambda_1(x) = \frac{2}{x} J_1(x)$$

# Power Pattern: Airy Pattern



This power pattern is known as the [Airy pattern](#). The first null is at  $x = \pi u D / \lambda \sim 3.832$  or  $u = \sin \theta \sim 1.22 \lambda / D$ . Thus the full beam width is

$$\text{BWFN} = 2.44 \frac{\lambda}{D} \text{ rad} \sim 140^\circ \frac{\lambda}{D}$$

The FWHM or HPBW is

$$\text{HPBW} = 1.02 \frac{\lambda}{D} \text{ rad} \sim 58.4^\circ \frac{\lambda}{D}$$

# Circular Gaussian Illumination

A good approximation for “real” radio telescopes with practical feeds and circular apertures is a **circular Gaussian illumination pattern**. In this case, we will use  $(x, y)$  instead of  $(\varrho, \varphi)$  to avoid the complicated Hankel transform. For 1D, the normalized Gaussian function is

$$g(u) \equiv \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{u^2}{2\sigma^2}\right)$$

where  $\sigma$  is the rms radius defined by

$$\sigma \equiv \left[ \int_{-\infty}^{\infty} u^2 g(u) du \Big/ \int_{-\infty}^{\infty} g(u) du \right]^{1/2}$$

and “normalized” means  $\int_{-\infty}^{\infty} g(u) du = 1$ . For 2D, we have

$$\begin{aligned} g(u, v) &\equiv g(u)g(v) = \frac{1}{2\pi\sigma_u\sigma_v} \exp\left[-\left(\frac{u^2}{2\sigma_u^2} + \frac{v^2}{2\sigma_v^2}\right)\right] \\ &\propto \exp\left(-\frac{u^2}{2\sigma_u^2}\right) \exp\left(-\frac{v^2}{2\sigma_v^2}\right) \end{aligned}$$

# Power Pattern

With the rms radius of the illumination pattern in wavelength  $\sigma_u = \sigma_v = \sigma$ , the  $E$  field pattern is

$$f(l, m) \propto \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2\sigma^2}\right) \exp\left(-\frac{v^2}{2\sigma^2}\right) \exp[-i2\pi(lu + mv)] du dv$$

with  $u = x/\lambda$  and  $v = y/\lambda$ ,  $l = \sin \theta_x$  and  $m = \sin \theta_y$ .

The function  $f(l, m)$  is separable:

$$f(l, m) \propto f(l) \times f(m)$$

with

$$f(l) \equiv \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2\sigma^2}\right) \exp(-i2\pi lu) du$$

$$f(m) \equiv \int_{-\infty}^{\infty} \exp\left(-\frac{v^2}{2\sigma^2}\right) \exp(-i2\pi mv) dv$$

$\Rightarrow$  Fourier transform in each direction!

# Power Pattern

---

The Fourier transform of a Gaussian is also a Gaussian:

$$f(x) = \exp(-ax^2) \Rightarrow F(s) = \sqrt{\pi/a} \exp(-\pi^2 s^2 / a)$$

Comparing with previous slide, we have  $a = 1/(2\sigma^2)$ , and thus

$$f(l) \propto \exp(-2\pi^2 l^2 \sigma^2)$$

$$f(m) \propto \exp(-2\pi^2 m^2 \sigma^2)$$

So the field pattern is

$$f(l, m) \propto \exp[-2\pi^2 \sigma^2 (l^2 + m^2)]$$

and the normalized power pattern is

$$P_n(l, m) = [f(l, m)]^2 \propto \exp[-4\pi^2 \sigma^2 (l^2 + m^2)]$$

Thus, the power pattern produced by a circular Gaussian illumination is also a circular Gaussian.

# HPBW

---

Substituting  $l^2 + m^2 = \sin^2 \theta$ , we have

$$P_n(\theta) \propto \exp[-4\pi^2 \sigma^2 \sin^2 \theta]$$

The power is half when  $\theta = \theta_h$  with

$$\exp[-4\pi^2 \sigma^2 \sin^2 \theta_h] = \frac{1}{2}$$

or

$$\theta_h = \sin^{-1} \left( \frac{\sqrt{\ln(2)}}{2\pi\sigma} \right)$$

If  $\theta_h \ll 1$  radian, we have

$$\theta_h \sim \frac{\sqrt{\ln(2)}}{2\pi\sigma} \text{ rad}$$

Thus, the half-power beam width is

$$\theta_{\text{HPBW}} = 2\theta_h \sim \frac{\sqrt{\ln(2)}}{\pi\sigma} \text{ rad}$$

where  $\sigma$  is the rms radius of the illumination pattern in wavelengths.

# How to choose $\sigma$ for Gaussian illum.?

Previously, we assume that the aperture itself is infinite and the Gaussian extends to  $\pm\infty$ . In practice, the aperture (reflector) is finite and the value of  $\sigma$  is chosen so that  $g(u, v)$  is small (e.g., 15 or 20 dB) at the edge of the aperture. Compared with uniform illumination, tapering (or grading) the illumination this way:

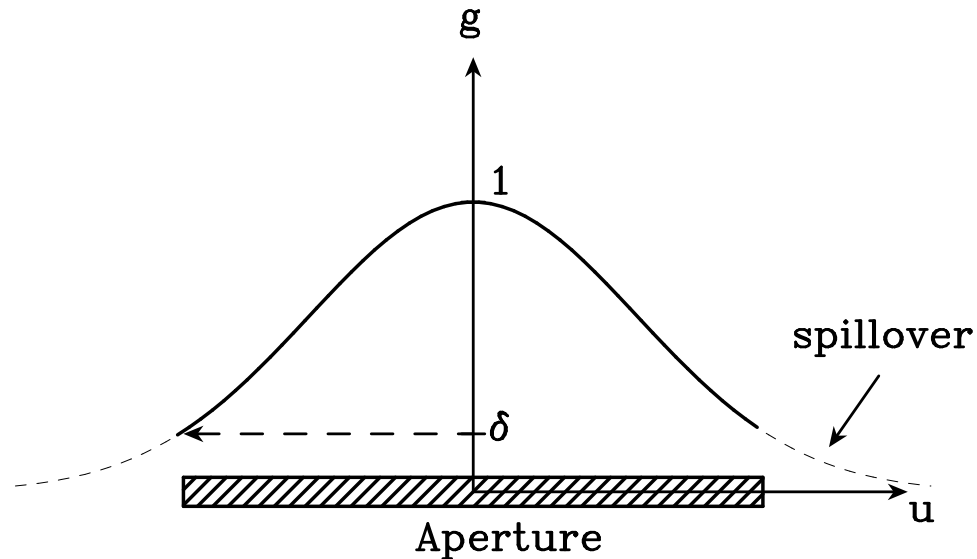
1. broadens the beamwidth  $\theta_{\text{HPBW}}$  slightly
2. reduces the aperture efficiency ( $\eta_A \propto A_e = \frac{D\lambda^2}{4\pi} \propto 1/\theta_{\text{HPBW}}^2$ ) slightly
3. reduces the sidelobe level significantly, and
4. reduces spillover (wasted power, noise from ground pickup due to reciprocity) significantly.

Here dB (decibel) is a measure of a ratio defined as

$$L_{\text{dB}} = -10 \log\left(\frac{P_1}{P_0}\right) \quad \Rightarrow \quad \frac{P_1}{P_0} = 10^{-\frac{L_{\text{dB}}}{10}}$$

with here  $P_0$  the power at the center (peak) and  $P_1$  at the edge.

# Spillover and Tapering



Spillover is illumination extending beyond the aperture. Here the relative field strength at the edge is  $\delta$  defined as (note  $P \propto E^2$ )

$$\delta \equiv \frac{g(u_{\max}, 0)}{g(0, 0)} \quad \Rightarrow \quad L_{\text{dB}} = -10 \log(\delta^2)$$

A field taper of  $\delta = 0.1$  corresponds to a power taper of  $\delta^2 = 0.01$  or a 20 dB edge taper. For an infinite aperture,  $\delta = 0$ . Note that the rms radius  $\sigma$  of the Gaussian illumination depends on and thus is set by  $\delta$ .

# Spillover and Tapering

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Let's find the rms radius  $\sigma$  of the Gaussian illumination for a given  $\delta$ . For a circular reflector with diameter  $D$ ,

$$u_{\max} = \frac{D}{2\lambda}$$

and

$$\delta = \frac{g(u_{\max}, 0)}{g(0, 0)} = \exp\left(-\frac{u_{\max}^2}{2\sigma^2}\right)$$

Then

$$-\ln(\delta) = \frac{u_{\max}^2}{2\sigma^2} = \frac{D^2}{8\lambda^2\sigma^2}$$

resulting in

$$\sigma = \frac{1}{\sqrt{-8\ln(\delta)}} \frac{D}{\lambda}$$

Recall that the beamwidth for  $\delta = 0$  is

$$\theta_{\text{HPBW}} = 2 \sin^{-1} \left( \frac{\sqrt{\ln(2)}}{2\pi\sigma} \right)$$

# HPBW

---

So for a given  $\delta$  and infinite aperture, we have

$$\theta_{\text{HPBW}} = 2 \sin^{-1} \left( \frac{\sqrt{-8 \ln(2) \ln(\delta)}}{2\pi} \frac{\lambda}{D} \right)$$

For example, if  $\lambda \ll D$ , then

$$\begin{aligned} \theta_{\text{HPBW}} &\sim 1.14 \frac{\lambda}{D} && \text{for } \delta = 0.10 \quad (20\text{dB}) \\ &\sim 0.98 \frac{\lambda}{D} && \text{for } \delta = 0.18 \quad (15\text{dB}) \end{aligned}$$

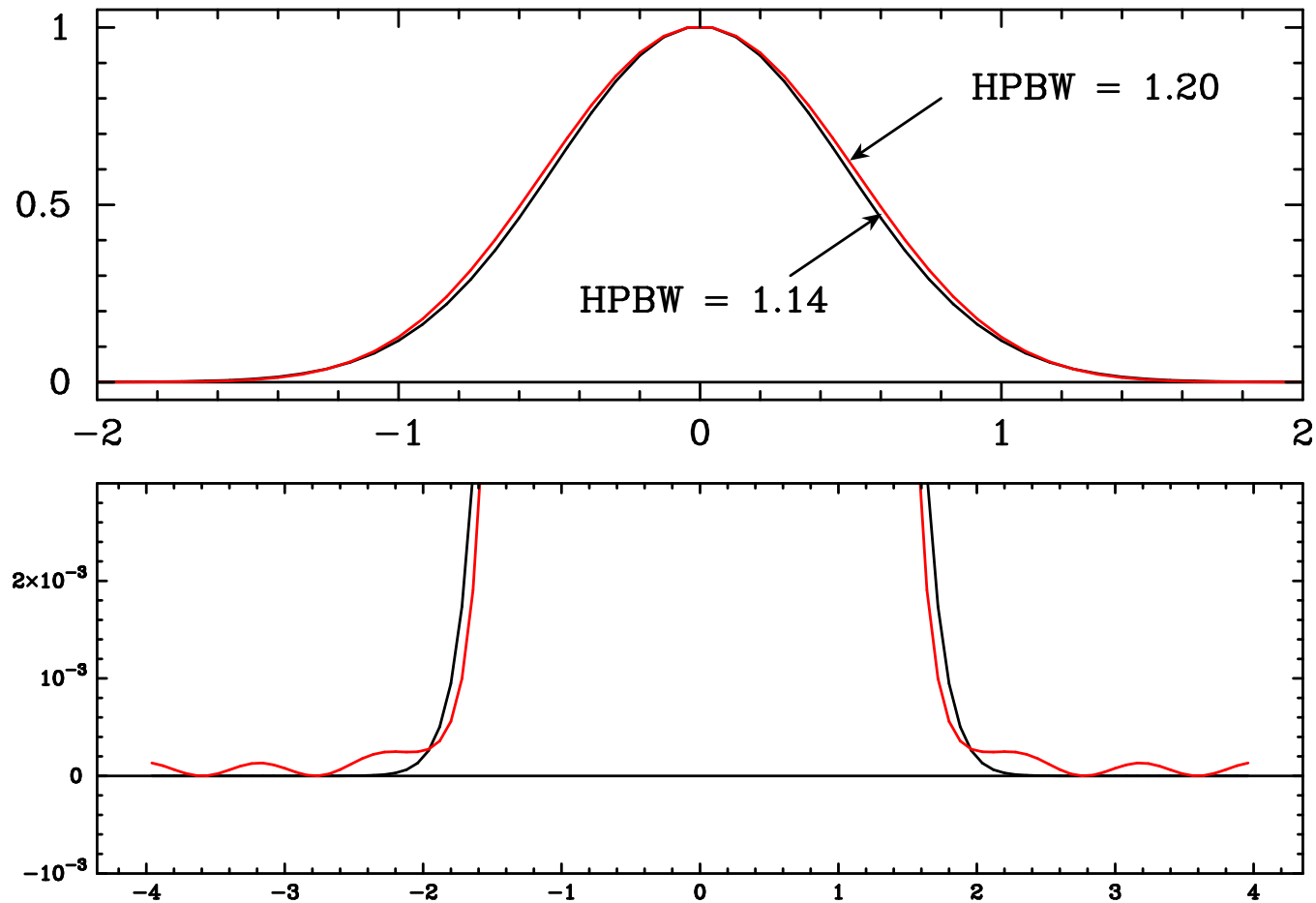
These estimates are slightly underestimated (e.g., by  $\sim 5\%$ ), as expected (see next slide) because infinite aperture width was used to derive the HPBW. A good rule of thumb for most single-dish radio telescopes is

$$\boxed{\theta_{\text{HPBW}} \approx 1.2 \frac{\lambda}{D}}$$

For example, the GBT ( $D = 100$  m), we have with  $\lambda = c/\nu$

$$\theta_{\text{HPBW}} \approx 1.2 \frac{c}{D\nu} \approx \frac{743 \text{ arcsec}}{\nu(\text{GHz})}$$

# Power Patterns for Gaussian Illumin.



Power patterns for Gaussian Illuminations with  $\delta = 0.1$ , with infinite (black) and finite (red) apertures.

# Reflector Accuracy Requirements

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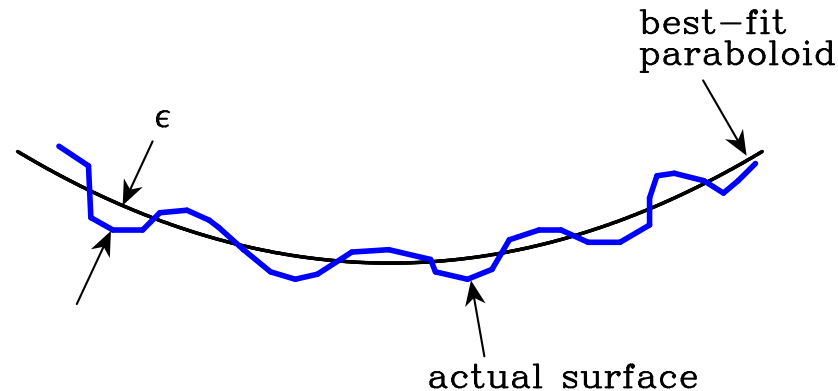
Real radio telescopes don't have perfectly paraboloidal reflectors. Small deviations from the best-fit paraboloidal may be caused by, e.g.,

1. permanent manufacturing errors
2. changing gravitational deformations as the reflector is tilted
3. thermal distortions resulting from solar heating
4. bending by strong winds

There will be some short wavelength  $\lambda$  below which these errors degrade the reflector performance so severely that the telescope becomes unusable.

Definition: the reflector surface efficiency  $\eta_s$  as the power gain of the actual reflector divided by the power gain of a perfectly paraboloidal reflector with the same size and illumination.

# Reflector Surface Efficiency



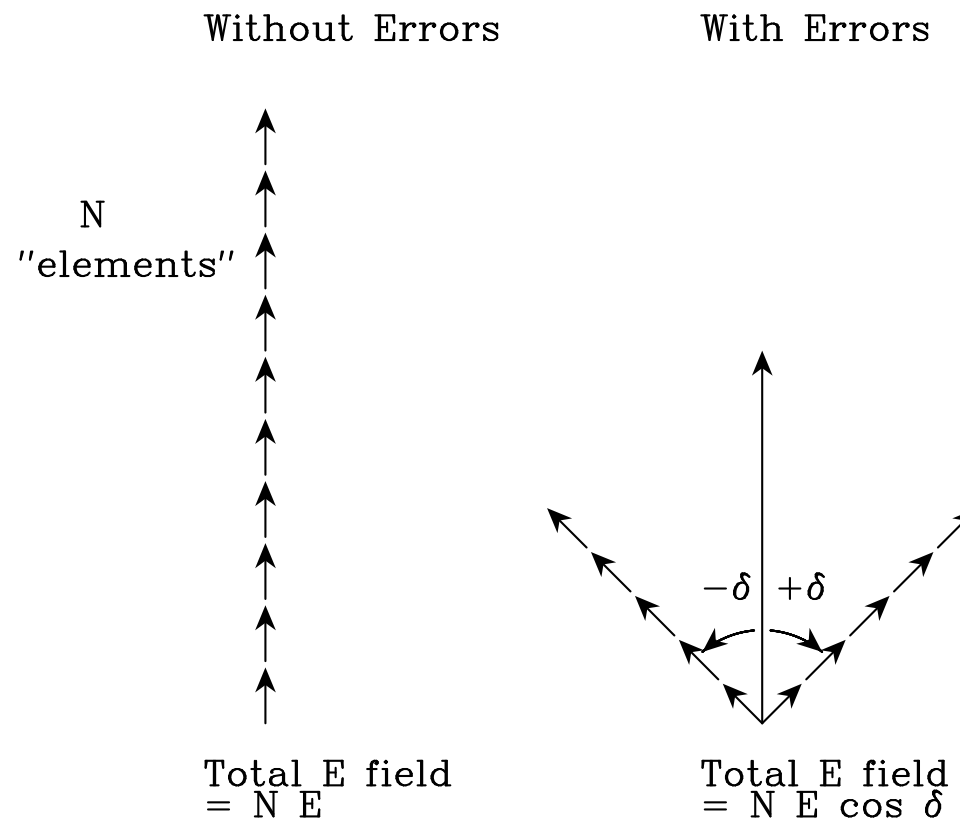
Where the actual reflector surface deviates from the best-fit paraboloid by a distance  $\epsilon$ , the path length of the reflected wave will be in error by almost  $2\epsilon$  and the phase error  $\delta$  (radians) will be

$$\delta = k(2\epsilon) = \frac{4\pi\epsilon}{\lambda}$$

This will cause a same phase error of the equivalent current density  $\mathbf{J}(\mathbf{x}')$  in the aperture plane.

# Reflector Surface Efficiency

An oversimplified example would be a bumpy surface, half covered with small bumps of height  $\epsilon \ll \lambda$  and half covered with small dips of the same depth  $\epsilon$ . Then the contribution of each area element to the far (electric) field is reduced by a factor  $\cos \delta$ :



Vectors sums of the electric fields produced by elements of perfect and imperfect apertures. Bumps in the imperfect aperture produce phase shifts  $\pm\delta$ .

# Reflector Surface Efficiency

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In the limits  $\delta \ll 1$  rad,  $\cos \delta \approx 1 - \frac{\delta^2}{2} \dots$  and

$$\frac{E(\delta)}{E(0)} = \cos \delta \approx 1 - \frac{\delta^2}{2} \dots$$

so the relative power gain is

$$\frac{G(\delta)}{G(0)} = \left[ \frac{E(\delta)}{E(0)} \right]^2 \approx 1 - \delta^2 \approx 1 - \left( \frac{4\pi\epsilon}{\lambda} \right)^2$$

This rough estimate shows that the surface errors must be an order-of-magnitude smaller than the shortest usable wavelength.

# Reflector Surface Efficiency

A more realistic calculation makes use of the fact that the most error distributions are roughly Gaussian. Suppose that the surface errors have a normalized Gaussian probability distribution  $P$  with rms  $\sigma$ :

$$P(\epsilon) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right)$$

Then the relative field strength is obtained as the weighted sum over all possible  $\epsilon$ :

$$\left\langle \frac{E}{E(0)} \right\rangle \approx \int_{-\infty}^{\infty} \cos\left(\frac{4\pi\epsilon}{\lambda}\right) \times \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) d\epsilon = \exp\left(-\frac{8\pi^2\sigma^2}{\lambda^2}\right)$$

Since Power  $\propto E^2$ , so our final result for the reflector surface efficiency is

$$\eta_s = \exp\left[-\left(\frac{4\pi\sigma}{\lambda}\right)^2\right]$$

which is often called the [Ruze equation](#). The aperture efficiency corrected for the error is

$$\eta_A = \eta_{A0}\eta_s = \eta_{A0} \exp\left[-\left(\frac{4\pi\sigma}{\lambda}\right)^2\right]$$

where  $\eta_{A0}$  is for an ideal surface. E.g. for the SMA,  $\sigma \sim 20\mu\text{m}$  (a fifth of the thickness of a sheet of paper!!!). So at  $\lambda \sim 1.3\text{ mm}$ , with  $\eta_{A0} \sim 0.78$ , we have  $\eta_A \sim 0.75$ .

# Reflector Surface Efficiency

Example: A traditional criterion for the shortest wavelength  $\lambda_{\min}$  at which a radio telescope works reasonably well is

$$\sigma = \frac{\lambda_{\min}}{16}$$

because the surface efficiency is only

$$\eta_s = \exp[-(\frac{\pi}{4})^2] \approx 0.54$$

and falling exponentially at shorter wavelengths.

The 100 m diameter GBT is intended to operate at frequencies as high as  $\nu \approx 100$  GHz, or  $\lambda_{\min} \approx 3$  mm. To meet this specification, the rms deviation from a perfect paraboloid must not exceed  $\sigma \approx 3\text{mm}/16 \approx 200 \mu\text{m}$ , the thickness of two sheets of paper!

$$G \propto 1/\lambda^2 \exp[-(\frac{4\pi\sigma}{\lambda})^2]$$

The power gain ( $G$ ) of a perfect paraboloidal reflector is proportional to  $\nu^2$ . If the reflector surface has a Gaussian error distribution with rms  $\sigma$ , then its gain increases as  $\nu^2$  at low frequencies, reaches a maximum at  $\lambda = 4\pi\sigma$  and falls exponentially at higher frequencies.

# Sensitivity

If a (lossless) antenna is pointed at a source of known flux density  $S_\nu$  with an angular diameter that is small compared to the telescope beam (i.e. to receive all the flux), a power received at the receiver input terminals is

$$W_\nu d\nu = \frac{1}{2} A_e S_\nu d\nu = k T_A d\nu$$

Thus

$$T_A = \frac{A_e}{2k} S_\nu \equiv \Gamma S_\nu$$

with the **sensitivity** defined as

$$\Gamma = \frac{A_e}{2k} \quad (\text{K Jy}^{-1})$$

With  $A_e = \eta_A A_g = \eta_A \pi D^2 / 4$ , we have

$$\Gamma = \eta_A \frac{\pi D^2}{8k}$$

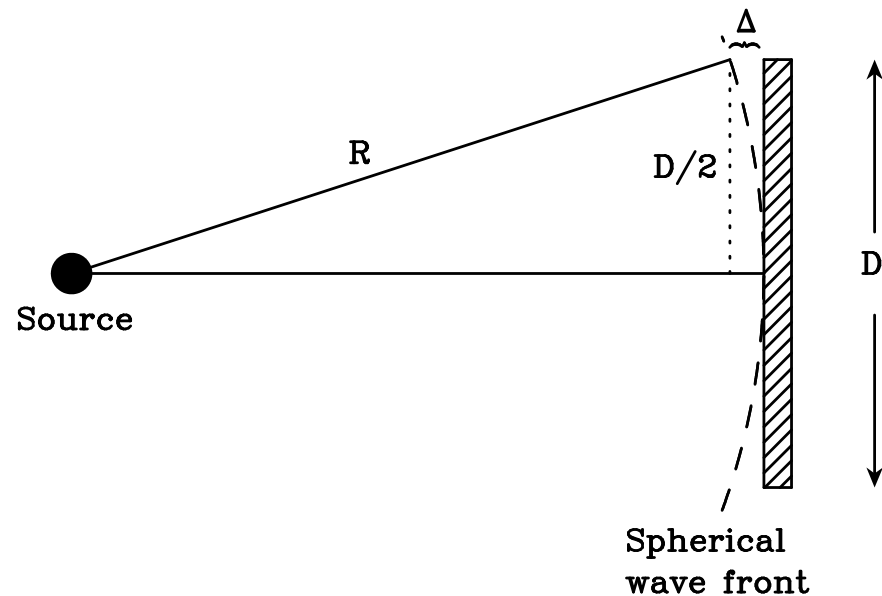
In practical work, the inverse of relation is often used because the noise depends on  $T_{\text{sys}}$  and is measured in  $T_A$ . Inserting numerical values we find

$$\Gamma^{-1} = \frac{3520}{\eta_A [D/\text{m}]^2} \quad \text{Jy/K}$$

E.g., for the SMA,  $\eta_A = 0.75$  at  $\lambda = 1.3\text{mm}$  and  $D = 6\text{m}$ , thus  $\Gamma^{-1} \sim 130 \text{ Jy/K}$ .

# The Far-field distance

How far away must a point source be for the received waves to satisfy our assumption that they are nearly planar? The answer depends on both the wavelength and the reflector diameter  $D$ . Consider the spherical wave emitted by a point source a distance  $R$  from the reflector:



The spherical wave emitted by a point source at distance  $R$  deviates from a plane by  $\Delta$  at the edge of an aperture whose size is  $D$ . The maximum departure  $\Delta$  from a plane wave occurs at the edge of the reflector. Conventionally, we define the far-field distance  $R_{\text{ff}}$  by requiring that  $\Delta < \lambda/2$ . Using the Pythagorean theorem, at the aperture edge

$$R^2 = (R - \Delta)^2 + (D/2)^2$$

# The Far-field distance

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Expanding the right side gives

$$R^2 = R^2 - 2R\Delta + \Delta^2 + D^2/4$$

and then

$$R = \frac{\Delta}{2} + \frac{D^2}{8\Delta} = \frac{D^2}{8\Delta} \left(1 + \frac{4\Delta^2}{D^2}\right) \approx \frac{D^2}{8\Delta}$$

assuming  $D \gg \lambda > 2\Delta$ . Substituting  $\Delta = \lambda/2$  at  $R = R_{\text{ff}}$  gives

$$\boxed{R_{\text{ff}} \approx \frac{D^2}{4\lambda}}$$

Unless  $R \gg R_{\text{ff}}$ , the path-length errors will introduce significant phase errors in the waves coming from the off-axis portions of the reflector, reducing the effective collecting area and degrading the antenna pattern.

Example: What is the far-field distance of the Green Bank Telescope ( $D = 100$  m) observing at  $\lambda = 1$  cm?  $R_{\text{ff}} \approx 250$  km.